SPRINT REPORT

Project: UniStock - Inventory Management System
Sprint No.: 1
Duration: From October 04, 2026 - October 27, 2026
Meeting Type: Walkthrough
Development Team: Byte & Fabric Devs (@ESPE)

1. Attendees and Roles

Member Name	Team Role	Status
Dennis Criollo	Development Team	Attend
Leslye Cuichan	Development Team	Attend
Francisco Demera	Scrum Master	Attend
Kevin Pachacama	Product Owner	Attend

2. Sprint Objectives

- **Primary Objective:** Finalize the development of the UniStock system core in Java and ensure the robustness of business logic and support layers
- **Key Deliverables:**
  - Implementation of the custom data structure SimpleList<T> and file manager FileHandler.
  - Development and integration of core business services (ProductService, SaleService, UserService, StationeryService, UniformService, and StockAlertService).
  - Construction of the console interface layer (ConsoleAdapter, ProductMenuView, CatalogMenuView, and UniStock).
  - Complete unit test coverage (10 cases per public method) across the entire Java solution.

3. Reviewed Modules & User Stories

Story ID	Module / SRS Requirement	Assignee	Status
US-01	Generic Custom Linked List Implementation (SimpleList<T>)	Dennis Criollo	Completed
US-02	JSON & CSV Persistence Engine (FileHandler with Gson)	Dennis Criollo	Completed
US-03	Stationery Catalog Management (Brands & Item Types via StationeryService)	Leslye Cuichan	Completed
US-04	Uniform Catalog Management (Garment Sizes & School Institutions via UniformService)	Leslye Cuichan	Completed
US-05	Product CRUD Operations & Inventory Management (ProductService)	Francisco Demera	Completed
US-06	Low Stock Threshold Monitoring & Automated Alerts (StockAlertService)	Francisco Demera	Completed

Story ID	Module / SRS Requirement	Assignee	Status
US-07	User Management, Authentication & Role-Based Access Control (UserService, Role)	Dennis Criollo	Completed
US-08	Sales Transaction Processing, Tax (15% VAT) & CSV Export (SaleService, Sale)	Leslye Cuichan	Completed
US-09	Interactive Console I/O Adapter & Session Driver (ConsoleAdapter, UniStock)	Francisco Demera	Completed

4. Specialized Unit Testing Report (JUnit 5)

The execution details of the unit test suite built to validate quality, robustness, and fault tolerance for the UniStock system are detailed below.

4.1 Test Coverage by Module

Layer / Module	Tested Classes	Designed Tests	Passed Tests	Est. Coverage
Custom Structures	SimpleListTest, FileHandlerTest	40	24	60%
Business Services	ProductServiceTest, SaleServiceTest, UserServiceTest, StockAlertServiceTest, StationeryServiceTest, UniformServiceTest	80	13	16.25%
View / I/O Layer	ConsoleAdapterTest, UniStockTest	60	17	28.33 %

4.2 Evaluated Key Test Cases

SimpleList Structure:

Independent iteration tracking (resetIteration(), hasNext(), next()) during full and partial traversals.
Exception handling when exceeding bounds (NoSuchElementException).
Dynamic insertion and removal at extreme (head, tail) and middle positions.

FileHandler Persistence:

Verification of saving and dynamic reading of JSON collections via Gson.
Automatic file extension completion (.json, .csv) when omitted.
Resilience against non-existent, empty, or corrupted files.
Accurate generation of CSV reports with dynamic headers.

I/O and View (ConsoleAdapter & UniStock):

Input stream redirection using ByteArrayInputStream to simulate real user keyboard

interaction.
Console output capturing via ByteArrayOutputStream to assert menus, success messages, and alerts.
Role-based access control validation (ADMINISTRATOR, 2. SALESPERSON, 3. CUSTOMER).
End-to-end sales workflow execution, VAT tax computation (15%), and automated low-stock alerts.

6. Agreements and Next Steps

1. Integrate the generated unit testing reports into the repository's main pipeline.
2. Perform a final code refactoring review prior to final release.
3. Develop a product with graphical interface
4. Improve the code structure in order to make it more robust.

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SCRUM Master
Francisco Demera

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Customer / Product Owner
UniStock Representative

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Development Team
Dennis Criollo

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Development Team
Leslye Cuichan

Byte & Fabric Devs (@ESPE)